

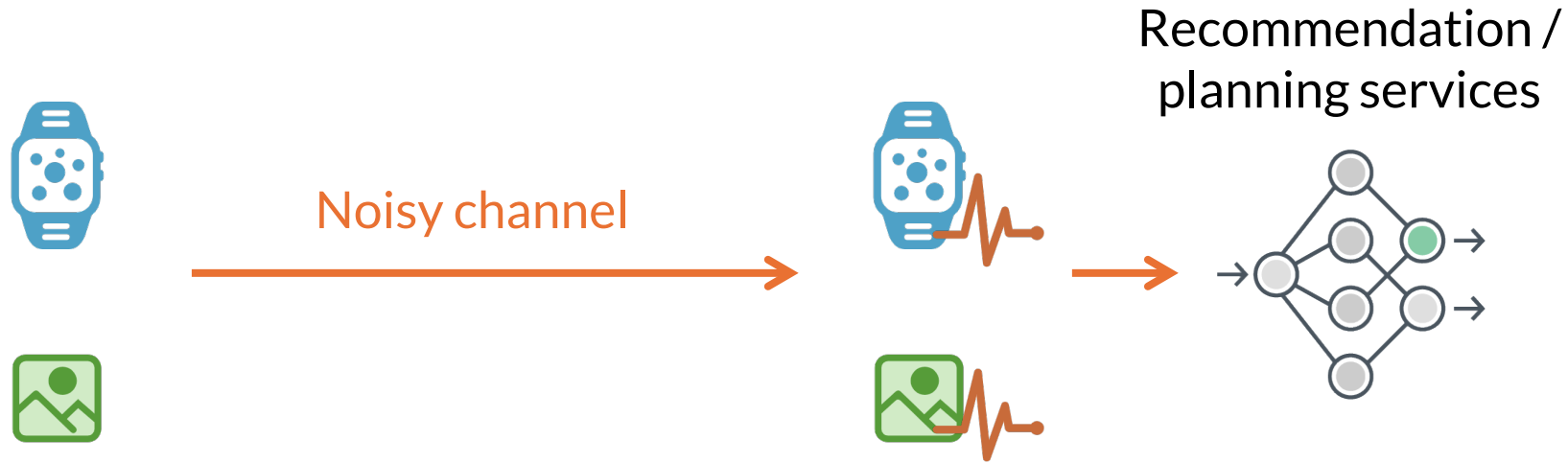
Quantifying Classifier Utility under Local Differential Privacy

Authors: Ye Zheng, Yidan Hu

RIT | Rochester Institute of Technology

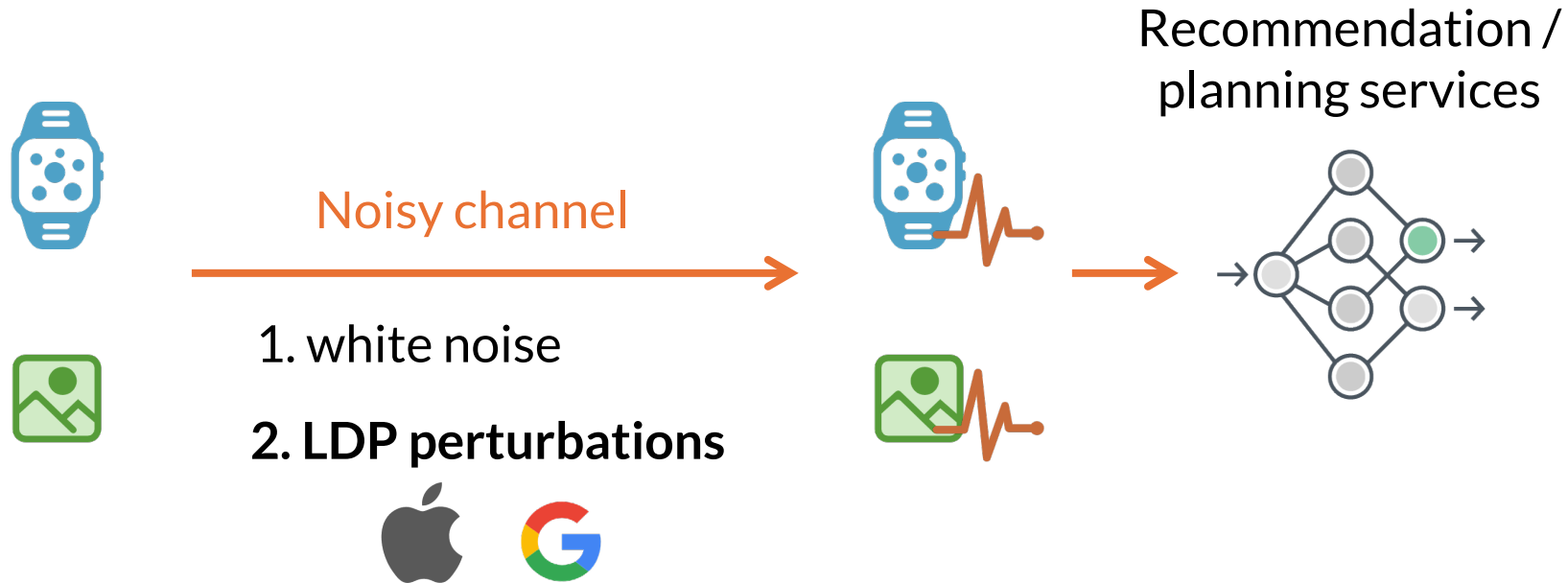
Classifier Utility under Noisy Inputs

- Input data for classifiers may be noisy



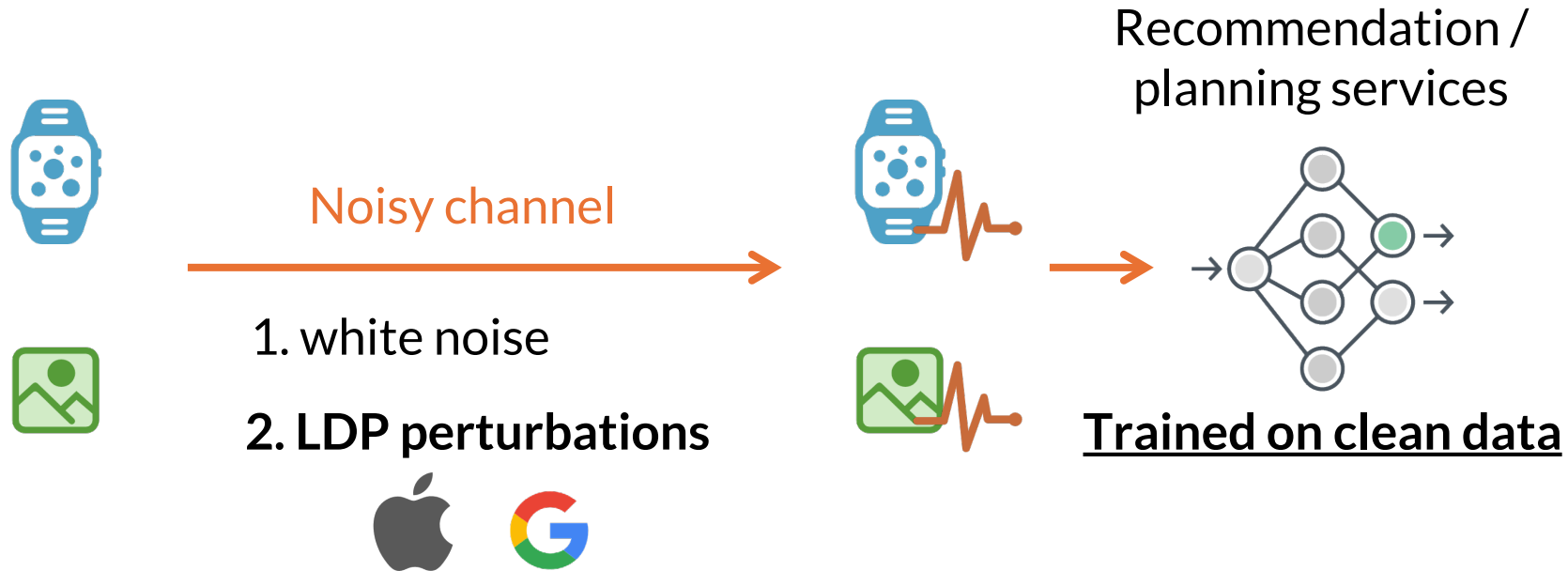
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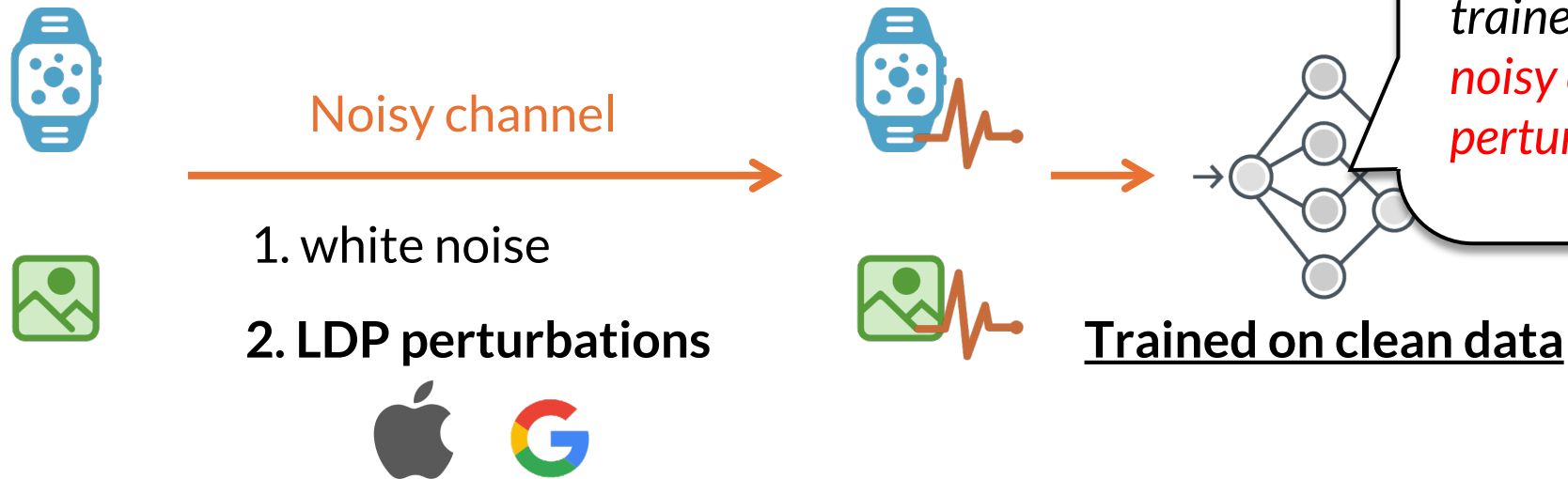
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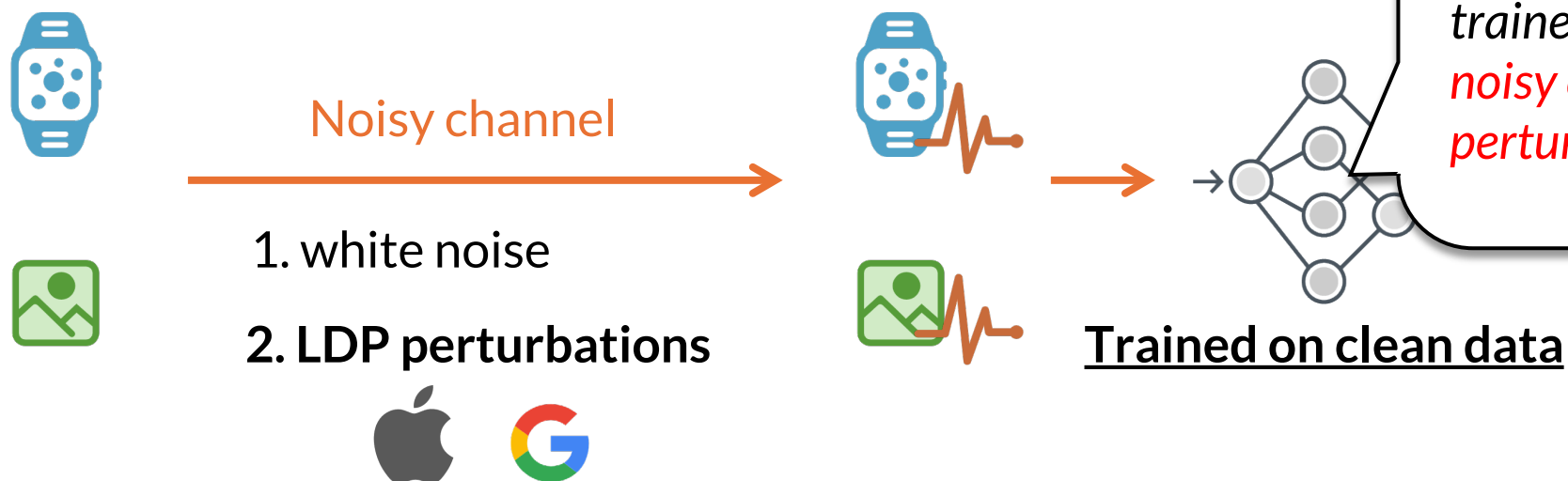
“

What's the utility of the trained classifier *under a noisy channel, e.g. LDP perturbation?*

”

Classifier Utility under Noisy Inputs

- Input data for classifiers may be noisy



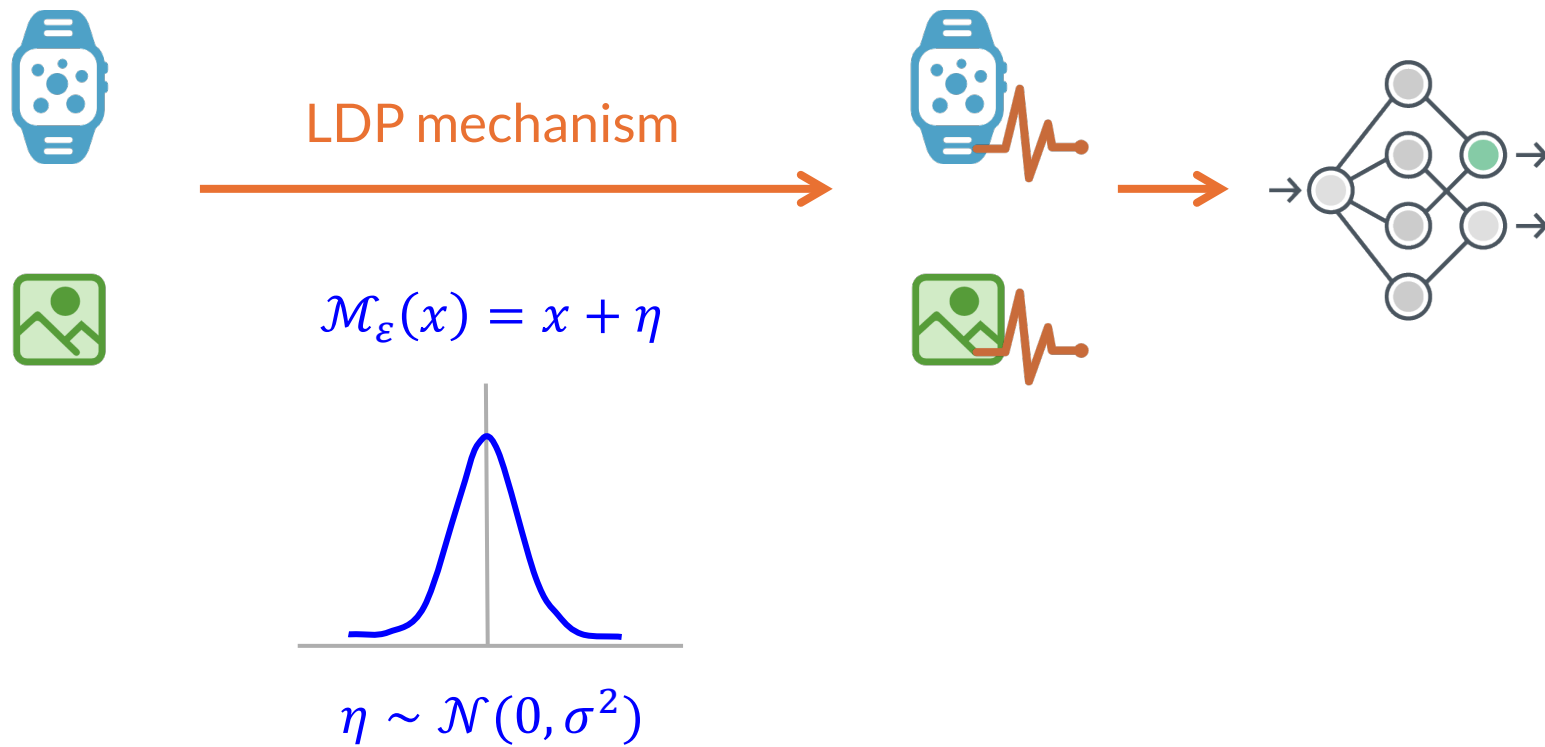
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For an LDP-friendly classifier

For a better privacy-utility balance

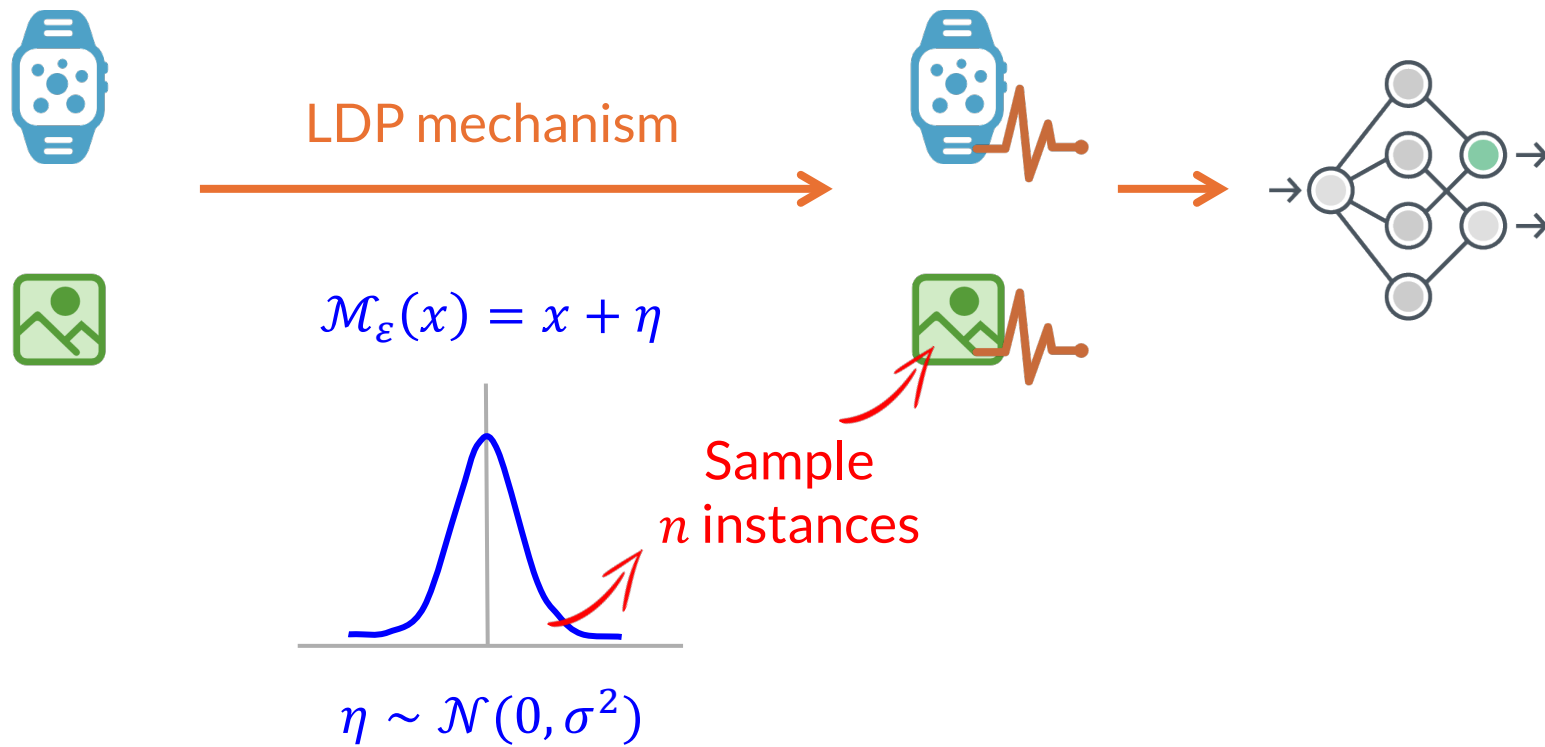
Empirical Classifier Utility under LDP-Data

- Empirical approach: Sample and then test
 - $\mathcal{M}$'s variance or MSE doesn't help – cannot provide a classifier accuracy



Empirical Classifier Utility under LDP-Data

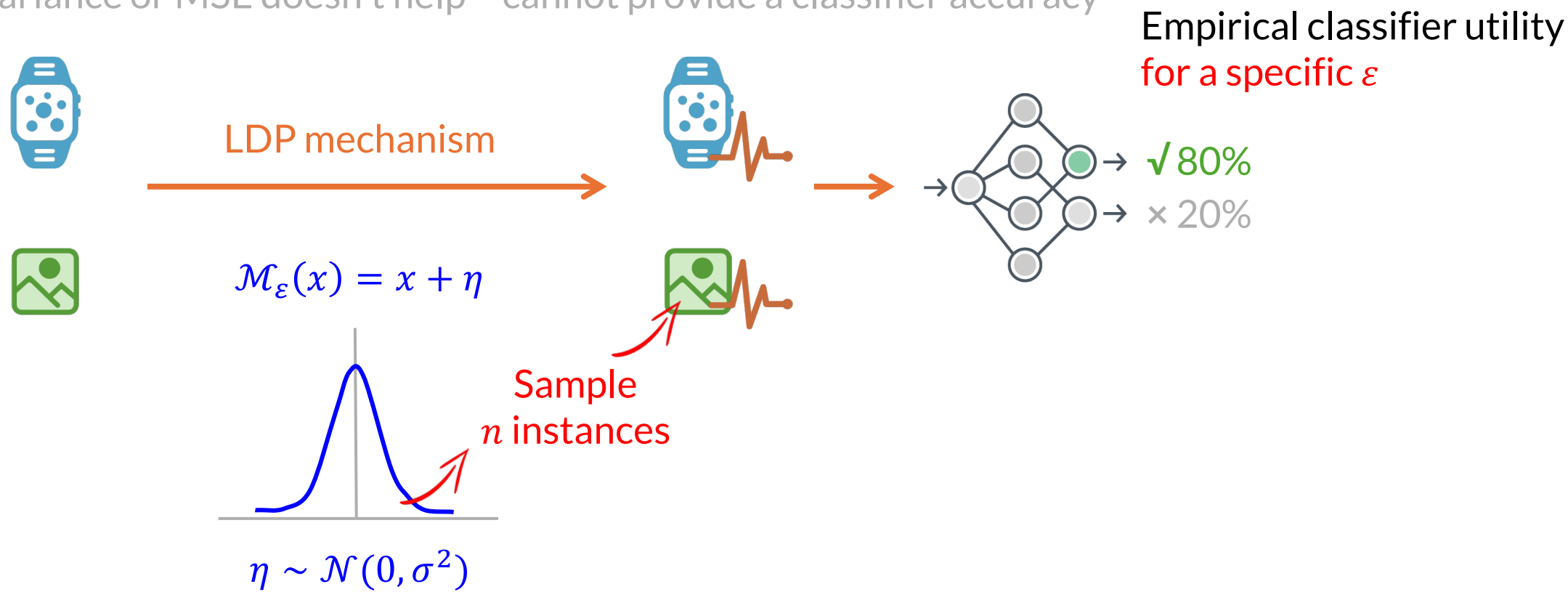
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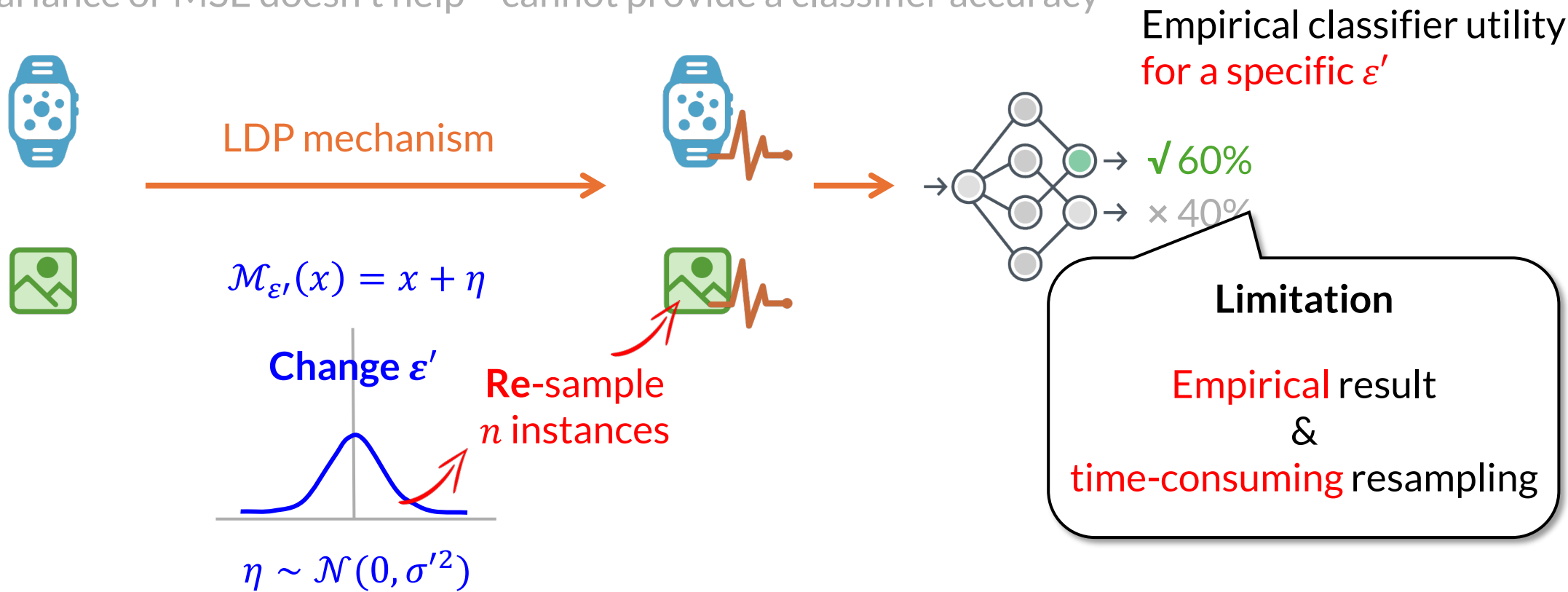
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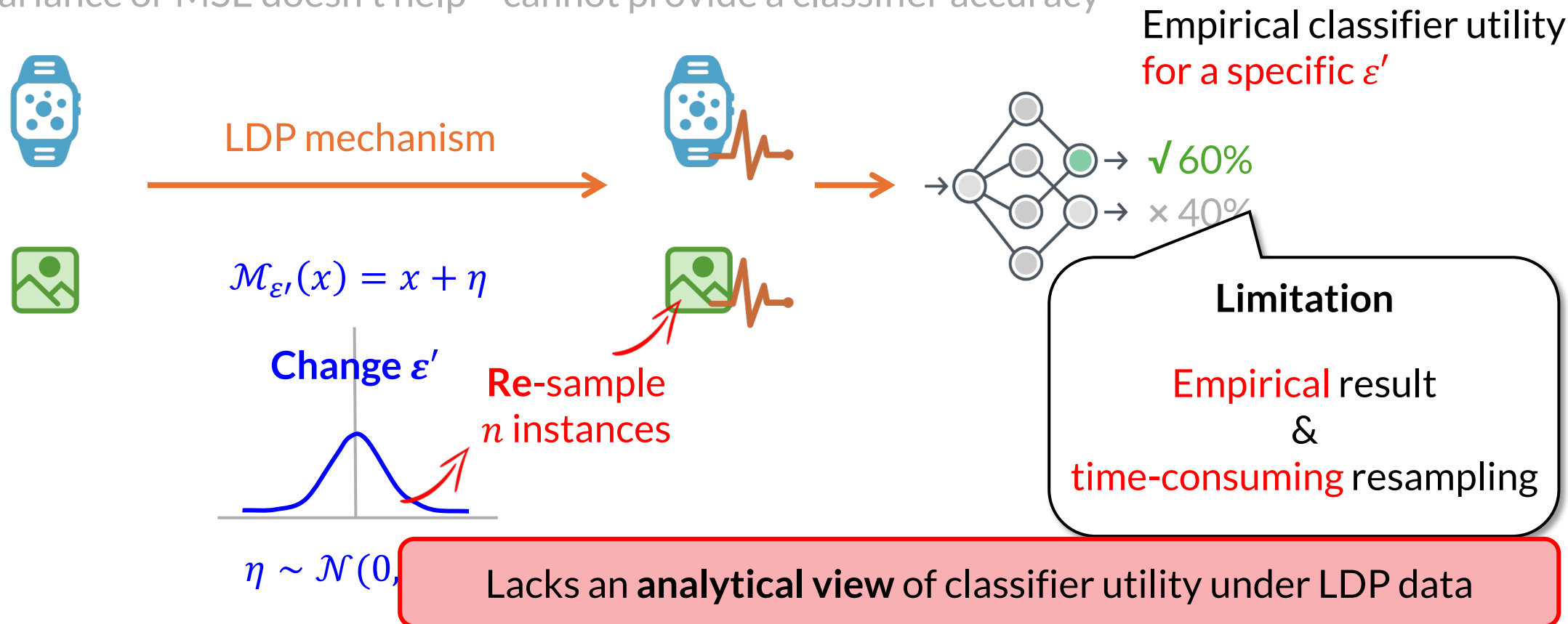
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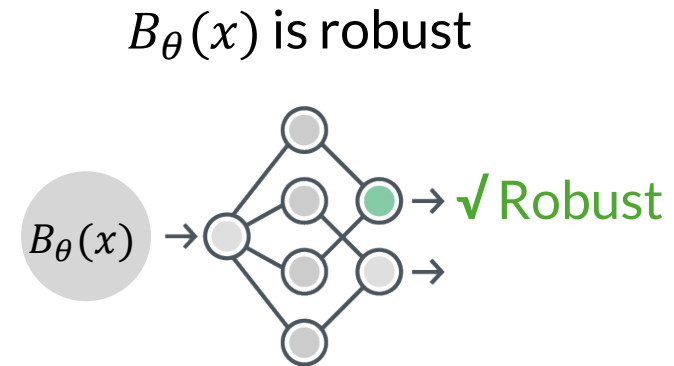
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Empirical Utility → Analytical Utility

- Analytical approach: **connecting LDP with robustness**

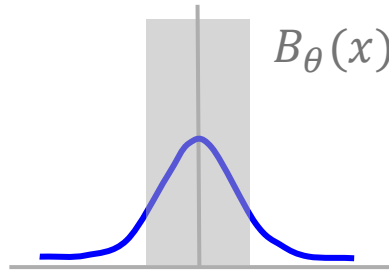


Robustness:
A θ -ball around x that doesn't
change classification

Empirical Utility → Analytical Utility

- Analytical approach: **connecting LDP with robustness**

$\mathcal{M}_\varepsilon(x)$ concentrates in $B_\theta(x)$

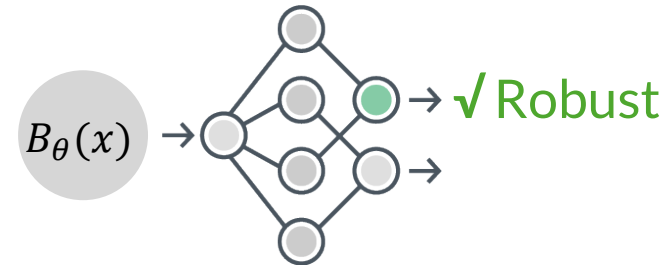


Concentration:

Probability of falling into a region

$$\Pr[\mathcal{M}_\varepsilon(x) \in B_\theta(x)] := p(\varepsilon, \theta)$$

$B_\theta(x)$ is robust

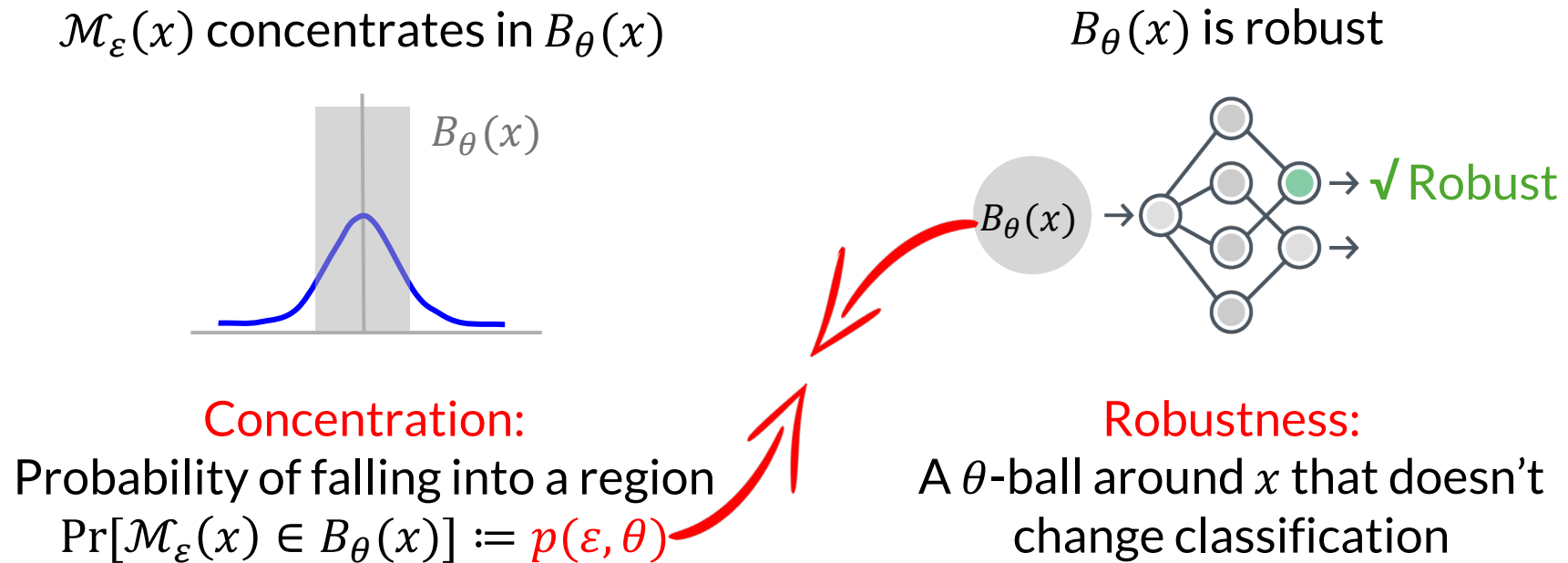


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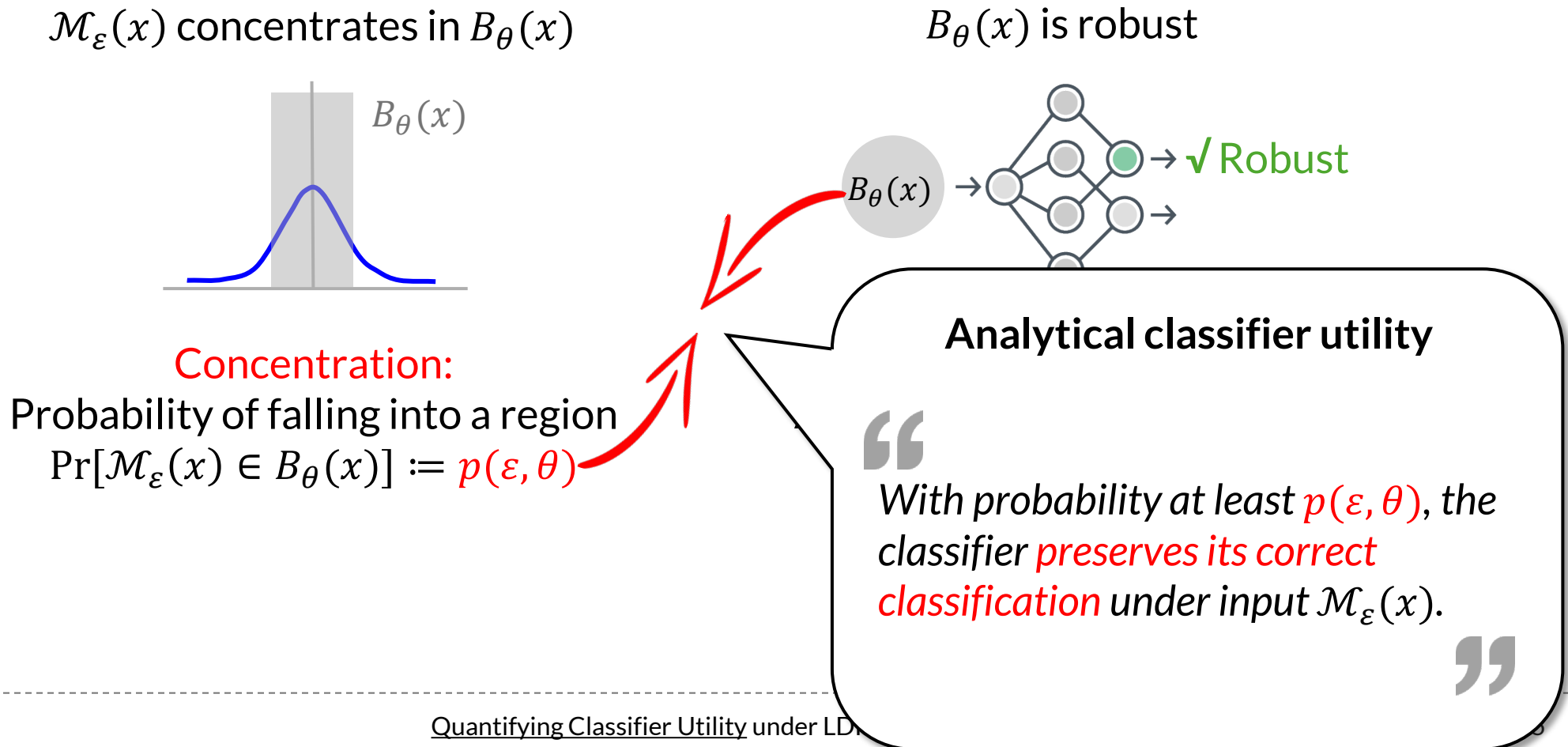
Empirical Utility $\rightarrow$ Analytical Utility

- Analytical approach: connecting LDP with robustness



Empirical Utility → Analytical Utility

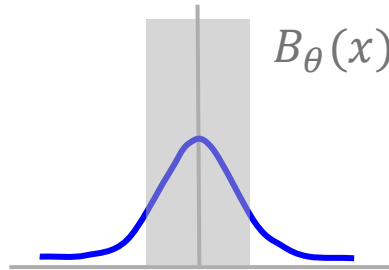
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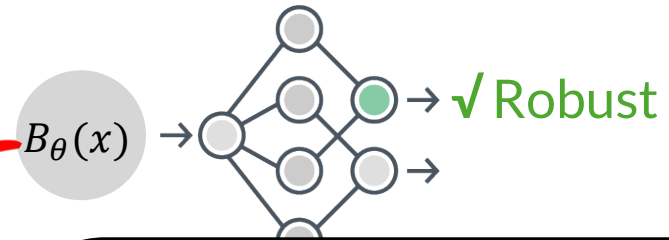


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$$\Pr[\mathcal{M}_\varepsilon(x) \in B_\theta(x)] := p(\varepsilon, \theta)$$

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Analytical classifier utility

“

With probability at least $p(\varepsilon, \theta)$, the classifier ~~preserves its correct~~ **classification** under input $\mathcal{M}_\varepsilon(x)$.

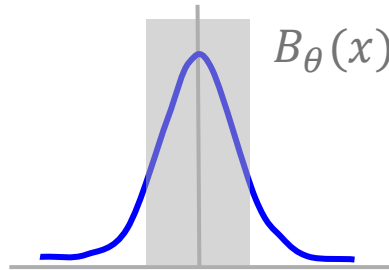
”

- Counterpart to the empirical **“80%”** accuracy

Empirical Utility → Analytical Utility

- Analytical approach: **connecting LDP with robustness**

$\mathcal{M}_{\epsilon'}(x)$ concentrates in $B_{\theta}(x)$

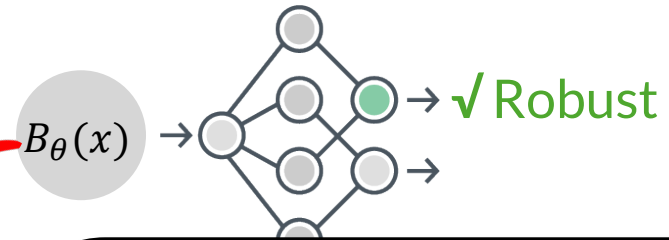


Concentration:

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$$\Pr[\mathcal{M}_{\epsilon'}(x) \in B_{\theta}(x)] := p(\epsilon', \theta)$$

$B_{\theta}(x)$ is robust



Analytical classifier utility

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With probability at least $p(\epsilon', \theta)$, the classifier **preserves its correct classification** under input $\mathcal{M}_{\epsilon'}(x)$.

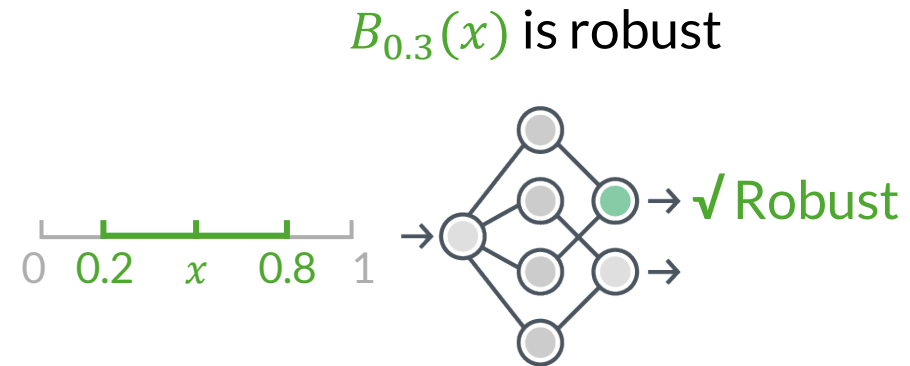
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Analytical & systematic view for any ϵ without resampling

One-Dimension Example

- Classifier $h: [0,1] \rightarrow \{1, 2\}$ under Laplace mechanism $\mathcal{M}_{\text{Lap}}(x)$

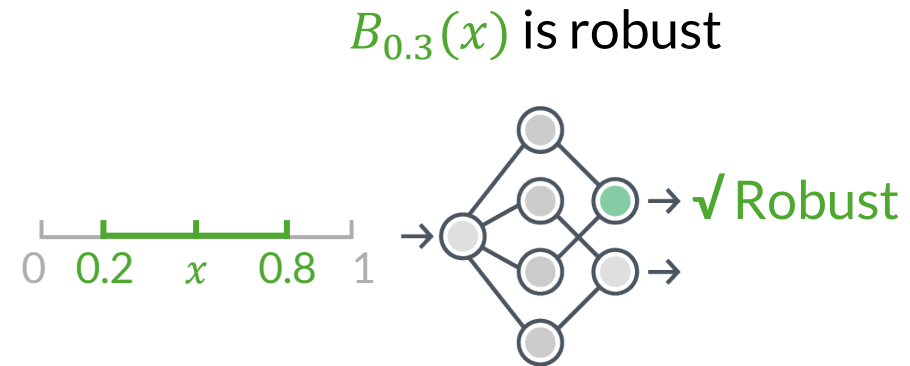
Classifier:
(robustness analysis)



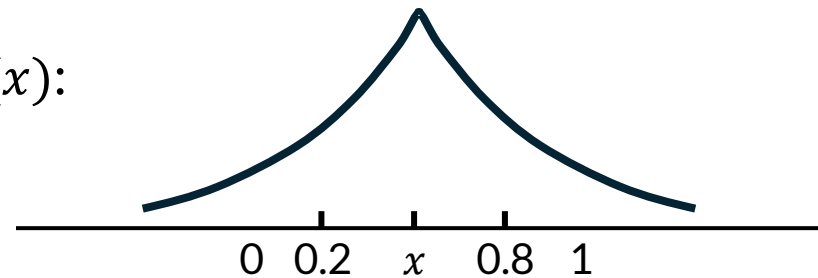
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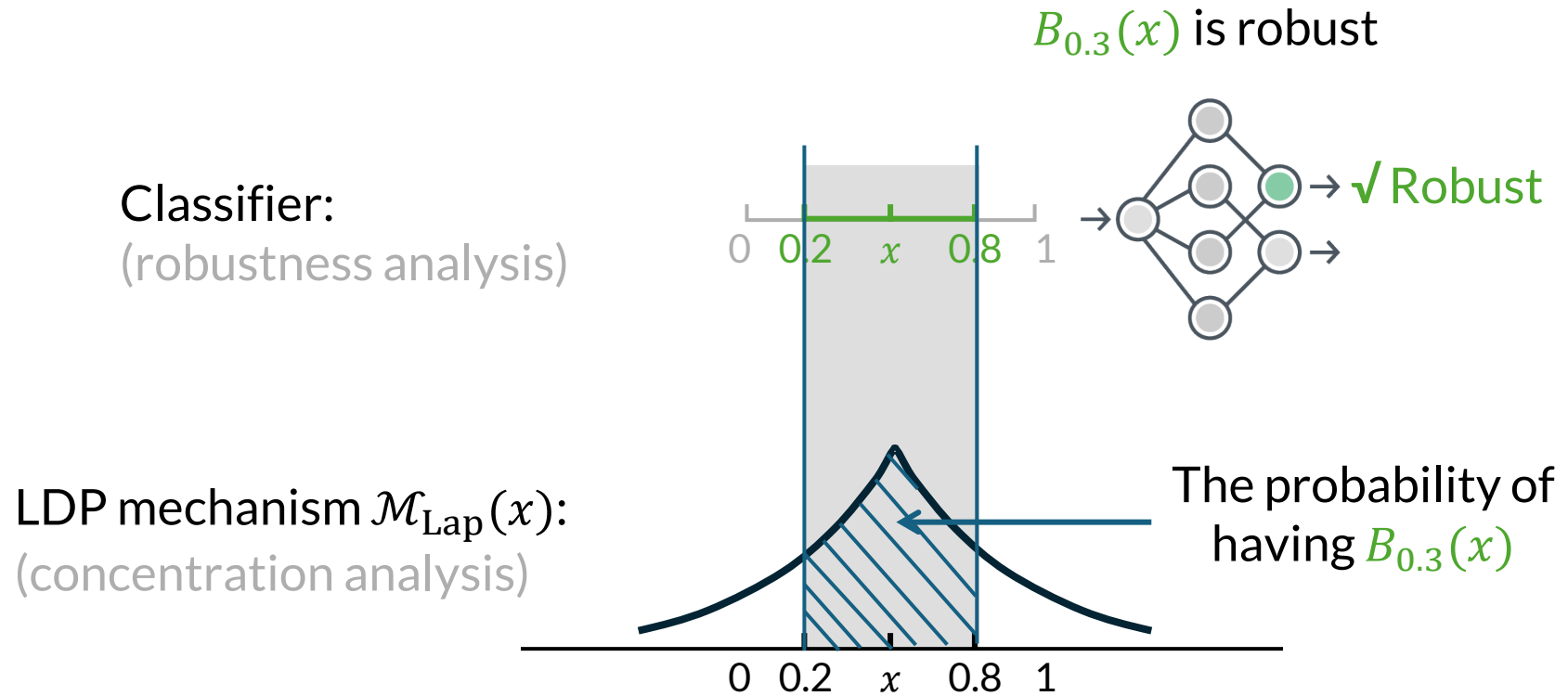


LDP mechanism $\mathcal{M}_{\text{Lap}}(x)$:



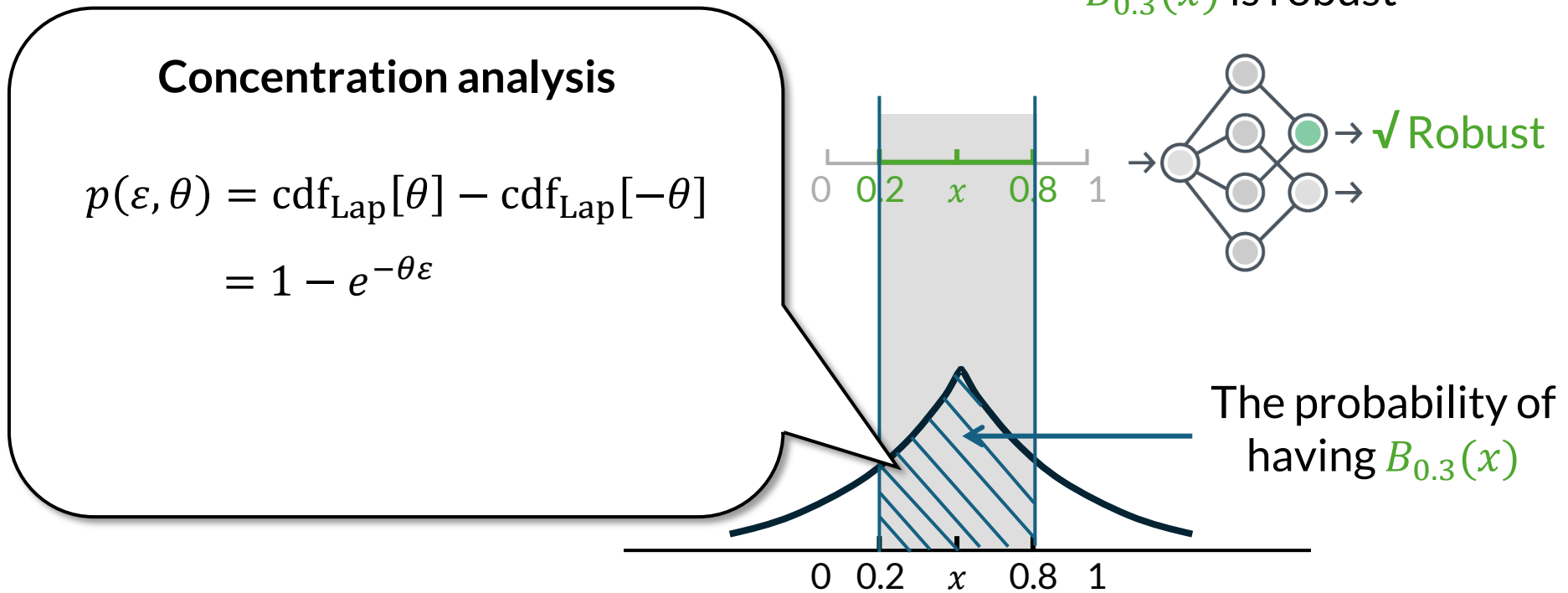
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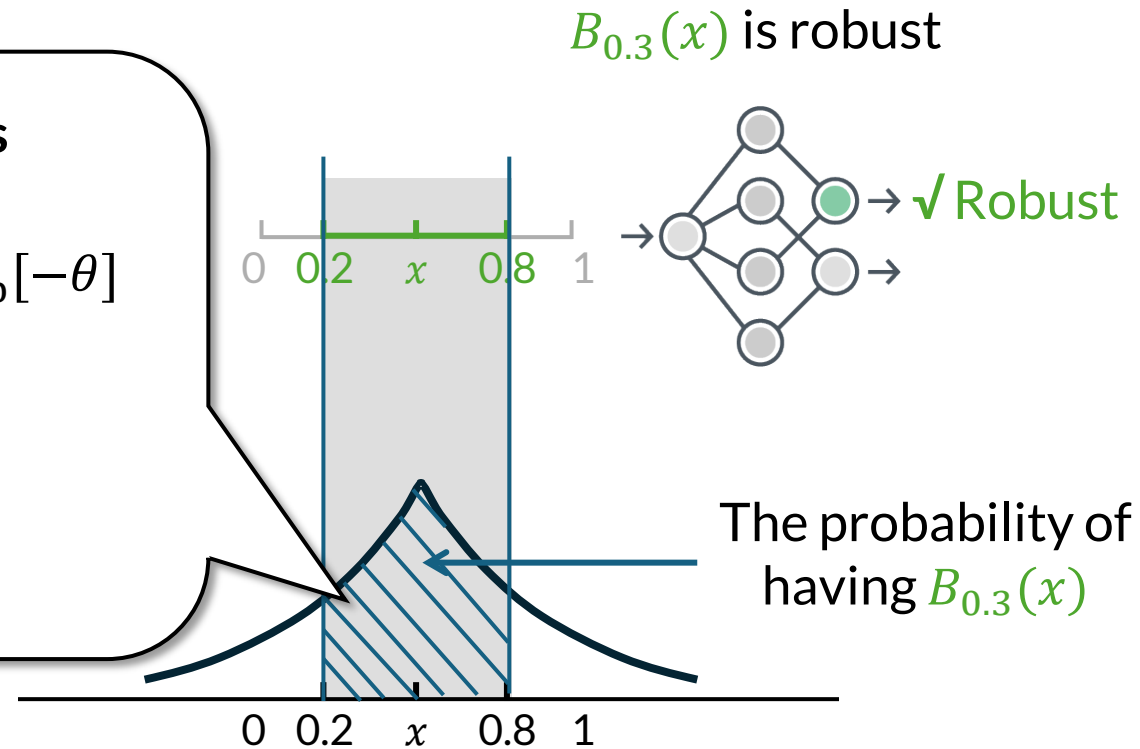
- Classifier $h: [0,1] \rightarrow \{1, 2\}$ under Laplace mechanism $\mathcal{M}_{\text{Lap}}(x)$

Concentration analysis

$$p(\varepsilon, \theta) = \text{cdf}_{\text{Lap}}[\theta] - \text{cdf}_{\text{Lap}}[-\theta]$$

$$= 1 - e^{-\theta\varepsilon}$$

$$p(\varepsilon = 2, \theta = 0.3) = 0.46$$



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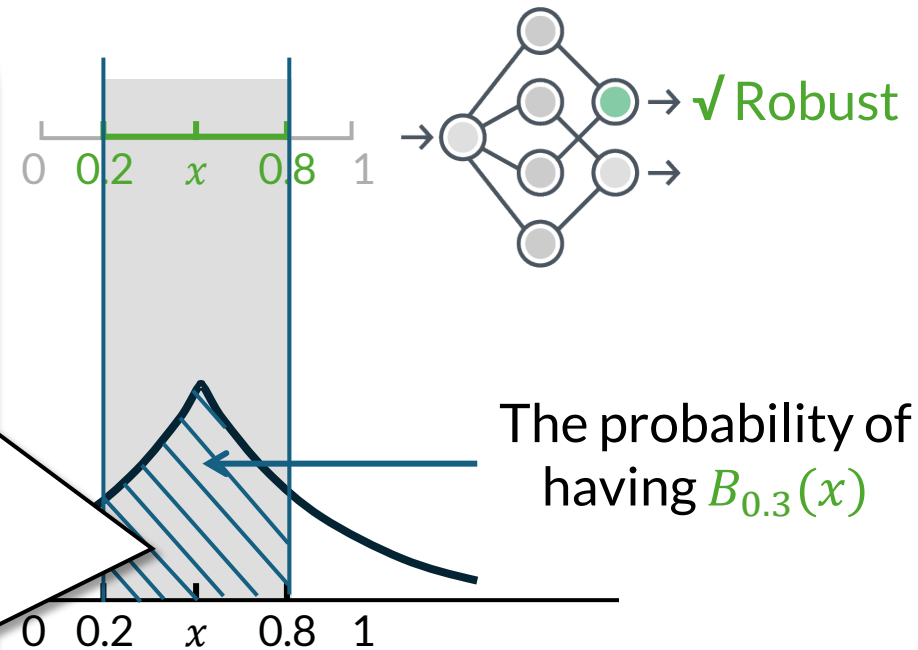
Concentration analysis

$$\begin{aligned} p(\varepsilon, \theta) &= \text{cdf}_{\text{Lap}}[\theta] - \text{cdf}_{\text{Lap}}[-\theta] \\ &= 1 - e^{-\theta\varepsilon} \end{aligned}$$

$$p(\varepsilon = 2, \theta = 0.3) = 0.46$$

“ With probability at least $p(2, 0.3) = 0.46$, the classifier **preserves its correct classification** under input $\mathcal{M}_{\varepsilon=2}(0.5)$. ”

$B_{0.3}(x)$ is robust



One-Dimension Example

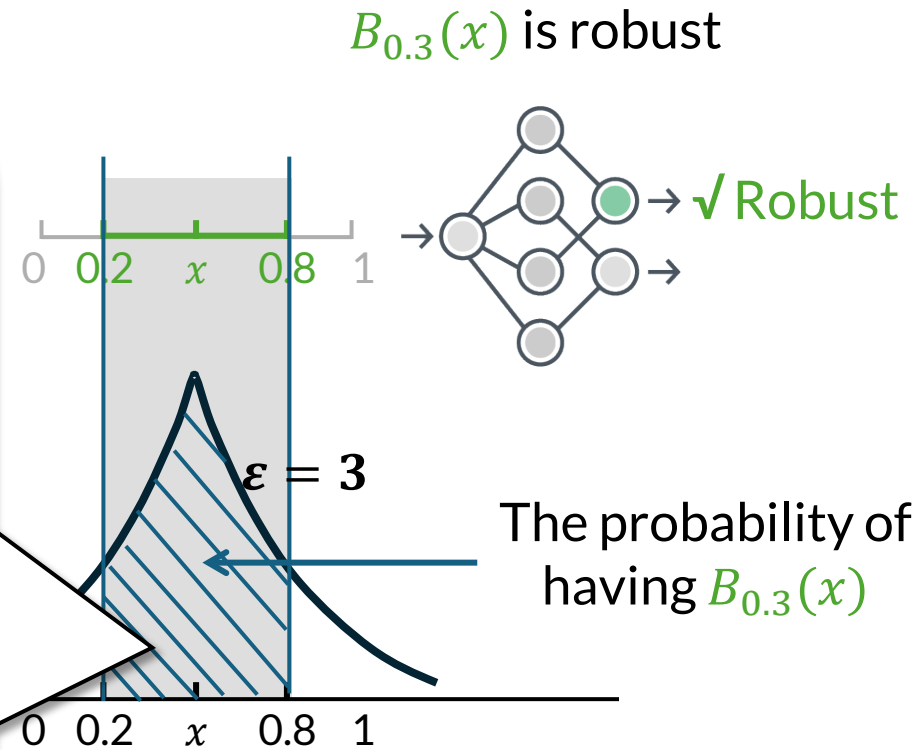
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For any ε :

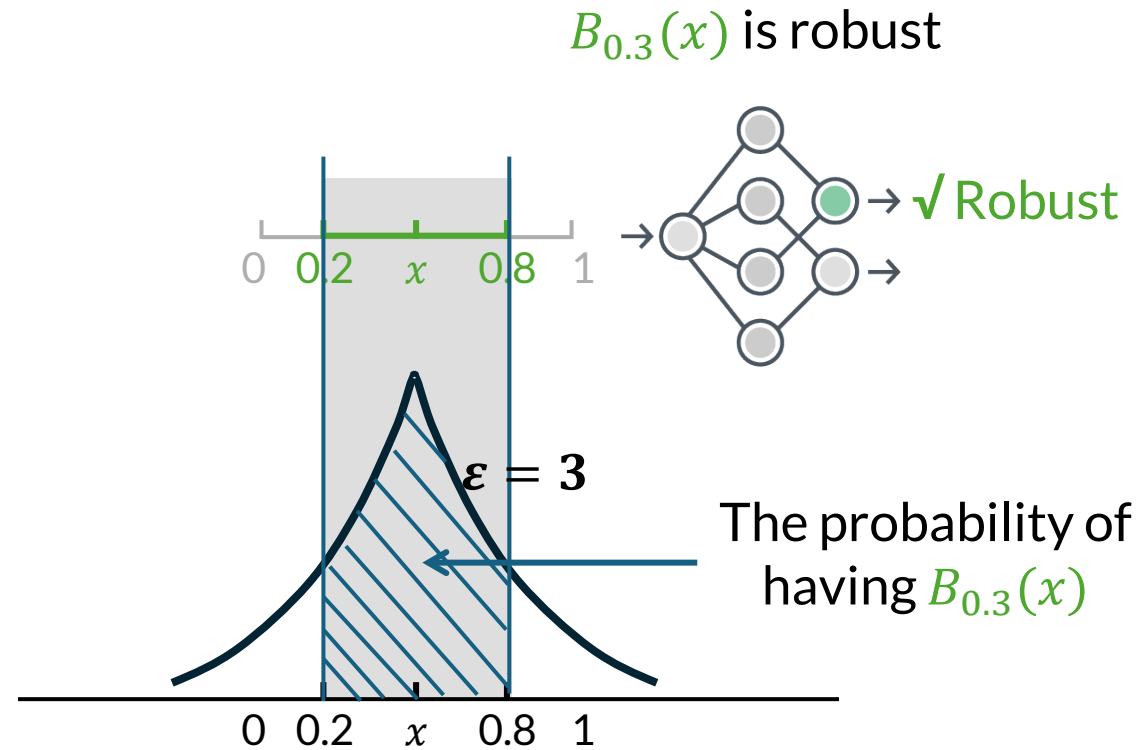
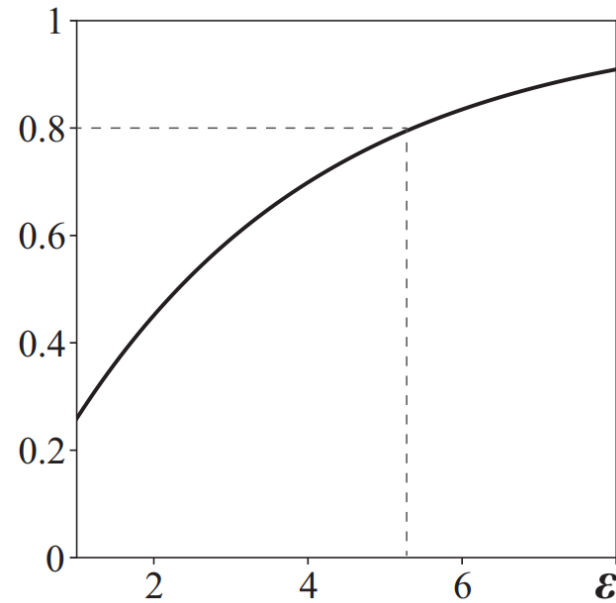
“ With probability at least $p(\varepsilon, \theta)$, the classifier **preserves its correct classification** under input $\mathcal{M}_{\varepsilon}(x)$. ”



Fixed Classifier (θ)

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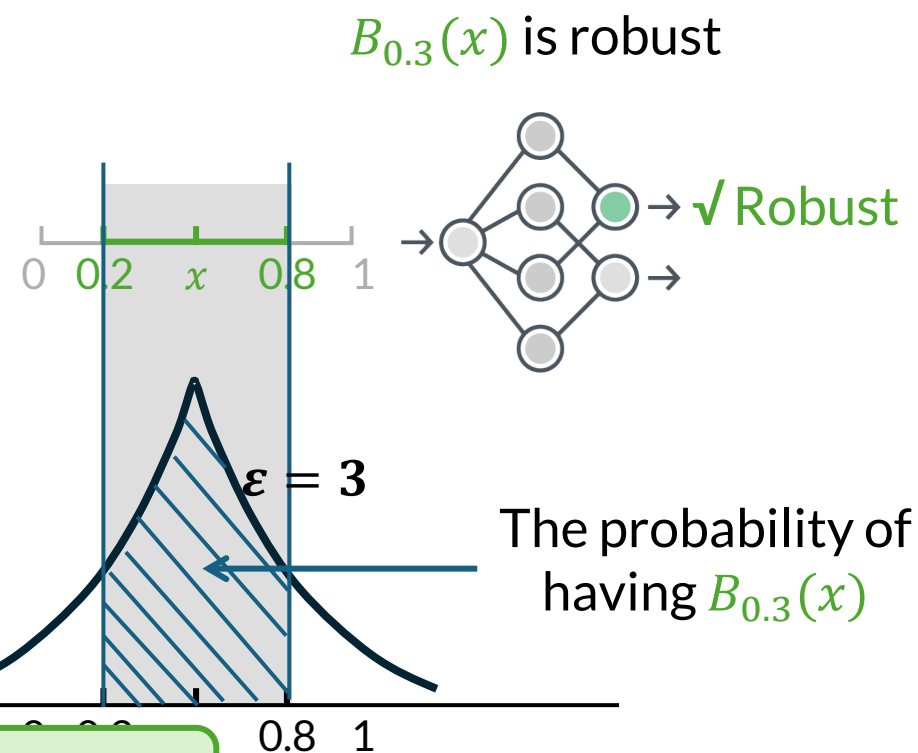
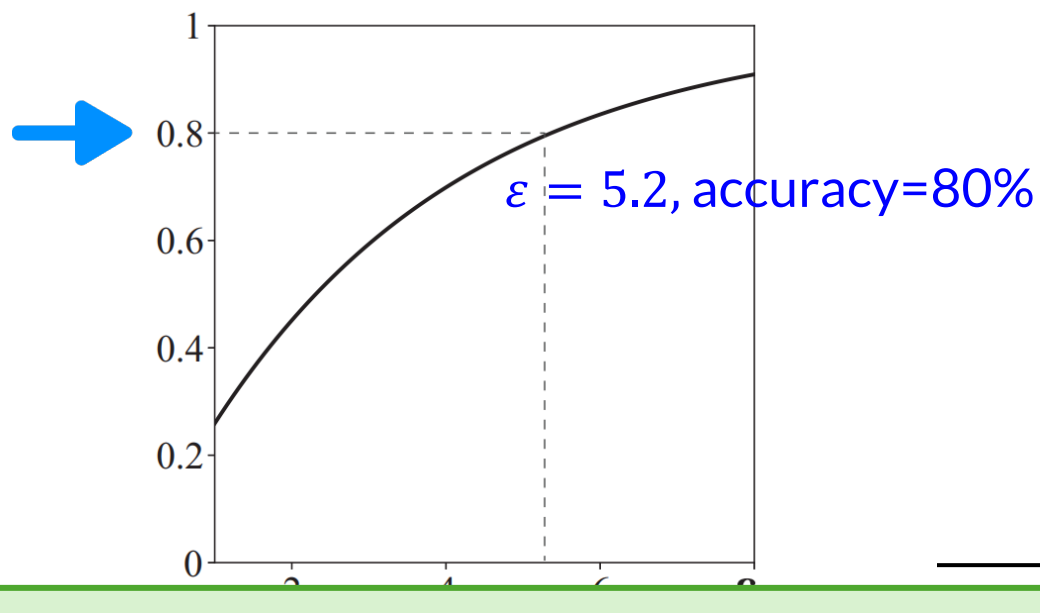
This classifier's utility $p(\epsilon, 0.3)$



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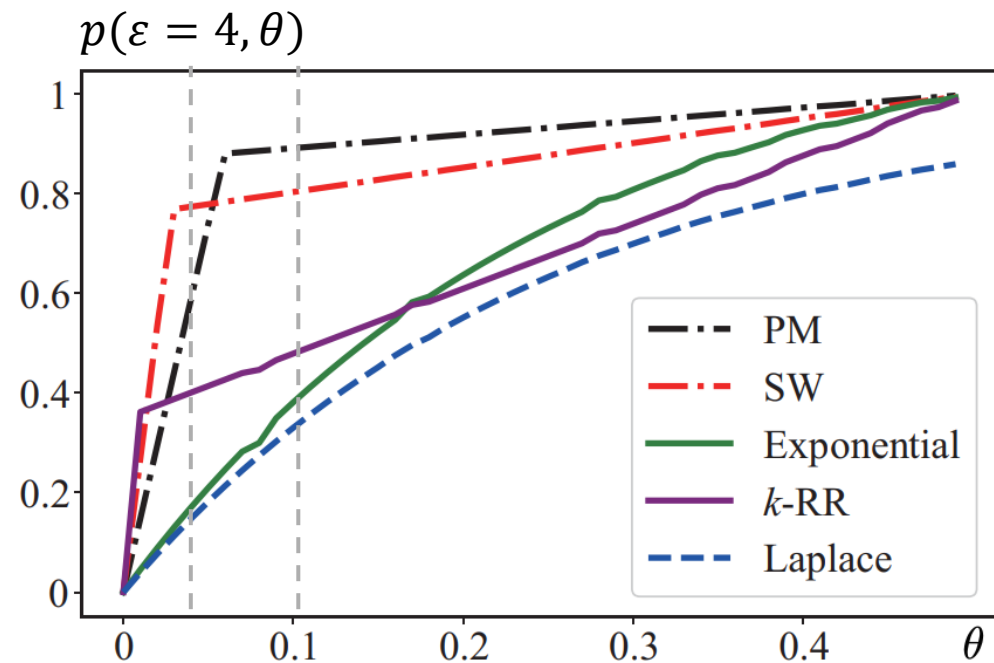
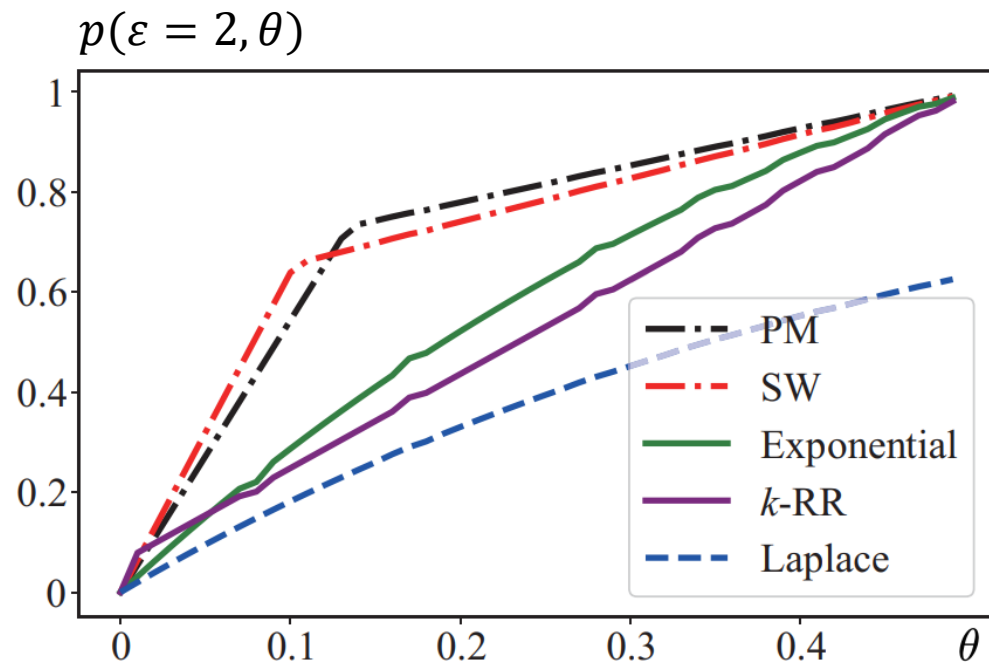
This classifier's utility $p(\epsilon, 0.3)$



Benefit 1: Choose the **best** ϵ for a desired classifier utility

Different $\mathcal{M}$ & Different Classifiers

- No universally optimal LDP mechanism for all ε and θ



Benefit 2: Systematic comparison of different LDP mechanisms

(b) $\varepsilon = 4$

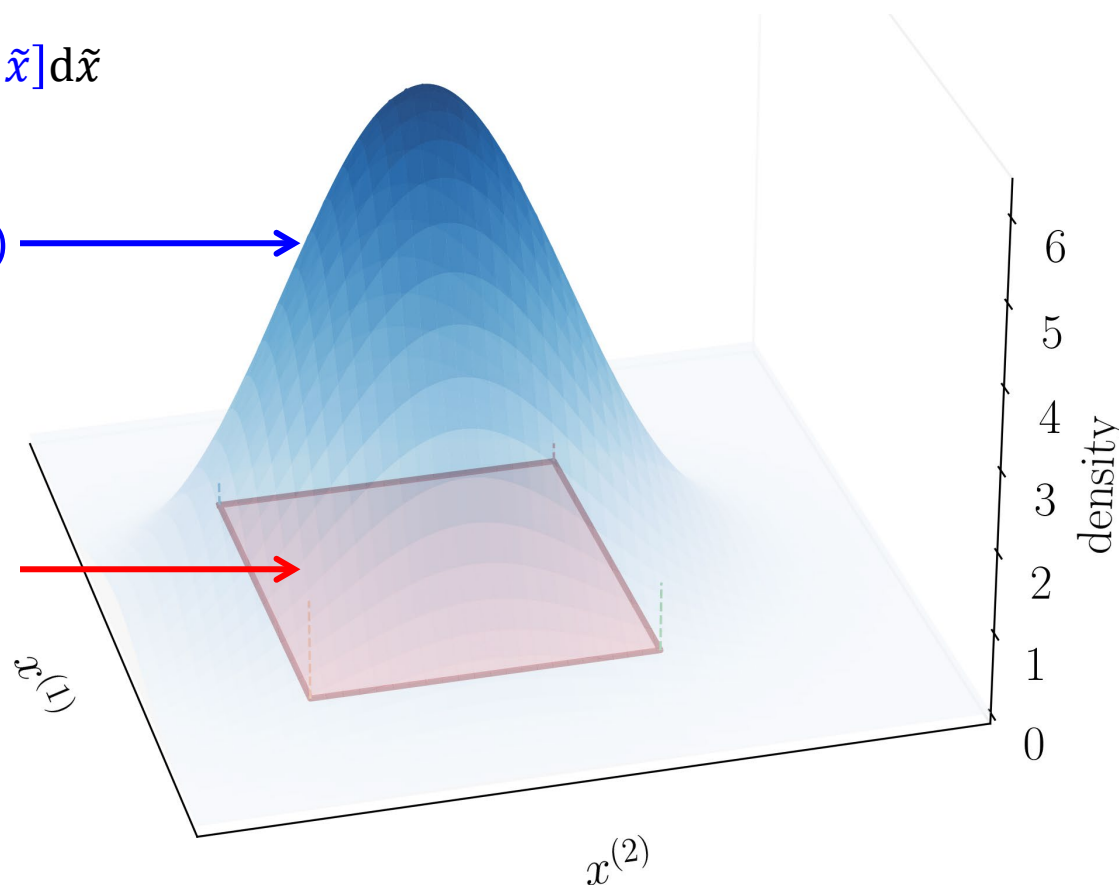
High-Dimensional Cases

- Concentration analysis on the robustness region S of a system

$$p(\varepsilon, S) = \int_S \text{pdf}[\mathcal{M}_\varepsilon(x) = \tilde{x}] d\tilde{x}$$

LDP mechanism $\mathcal{M}_\varepsilon(x)$ →

Robustness region S →



Case Studies – Quality of The Analytical Utility

- Claim: Accurate (compared with the empirical utility) & efficient

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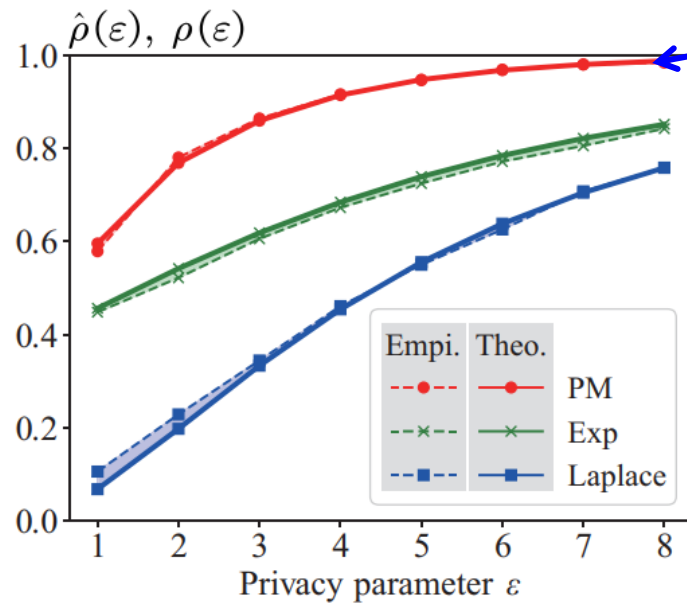
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 - Datasets: Stroke Prediction (medical data), Bank Customer Attrition (financial data), MNIST-7×7 (image data)
- 
- Diagram illustrating sensitive dimensions for datasets:
- Bank Customer Attrition (financial data) is associated with 2 sensitive dimensions.
 - MNIST-7×7 (image data) is associated with 49 sensitive dimensions.

Case Studies – Quality of The Analytical Utility

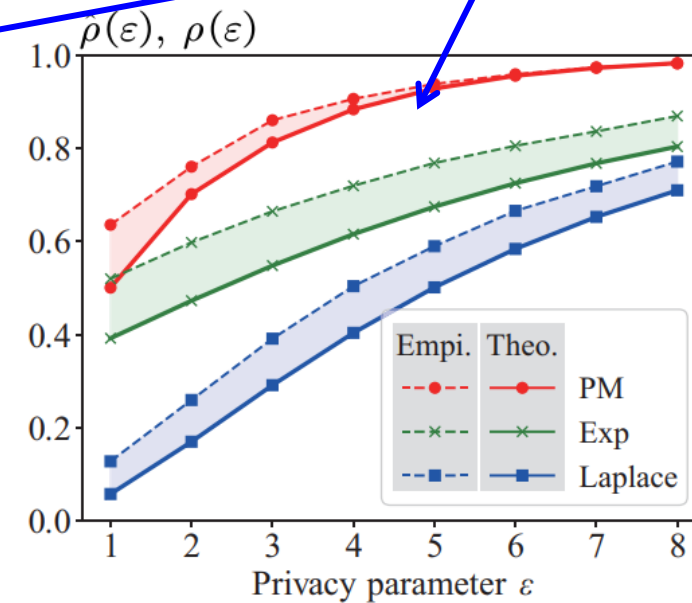
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 - low-dimensional: Logistic Regression, Random Forest
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- Datasets: Stroke Prediction (medical data), Bank Customer Attrition (financial data), MNIST-7×7 (image data)
- Classifier utility:
 - analytical utility $p(\varepsilon, S)$: approximated S for black-box classifiers
 - empirical utility $\hat{p}(\varepsilon)$: 2000 samples from $\mathcal{M}_\varepsilon$ and then used for testing

Case Studies – Stoke Predication

- x = the first record, with noisy “Age” and “BMI”



(a) Logistic Regression.



(b) Random Forest.

Case Studies – Stoke Predication

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Time cost comparison (ms)

	PM	Exponential	Laplace
Empirical^a	6.56 + 1.38	859.83 + 1.53	11.29 + 1.53
Theoretical^b	0.24	0.94	0.30

← Negligible time cost

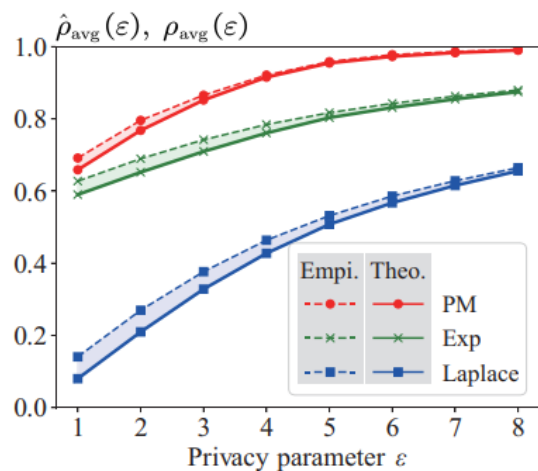
^a Time of 2000 samples + inference.

^b Time to compute $\rho(\varepsilon, S)$ only; computing S takes 5.80 ms but is a one-time cost amortized across all ε values.

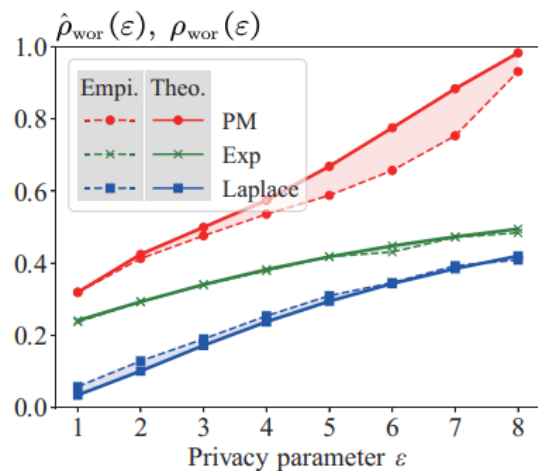
Case Studies – Stoke Predication

■ Average-case and worst-case utility

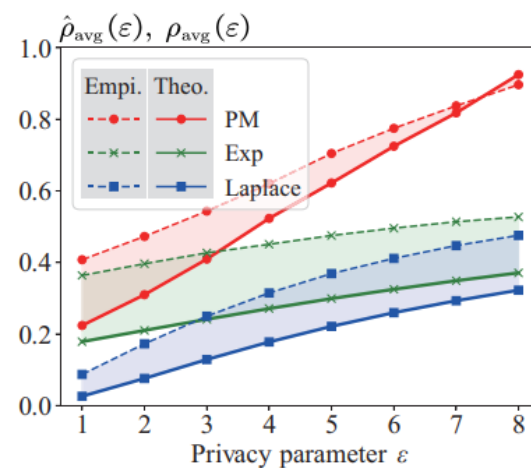
$$x \sim P_x \rightarrow \begin{cases} p_{\text{avg}}(\varepsilon) = \mathbb{E}_{x \sim P_x}[p_x(\varepsilon, S)] \\ p_{\text{wor}}(\varepsilon) = \min_{x \sim P_x}[p_x(\varepsilon, S)] \end{cases}$$



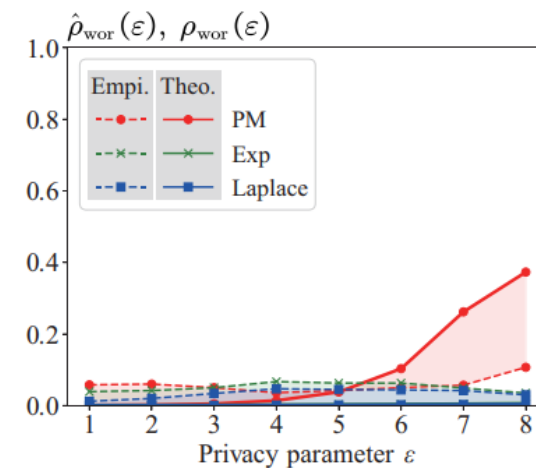
(a) LR: Average-case utility.



(b) LR: Worst-case utility.



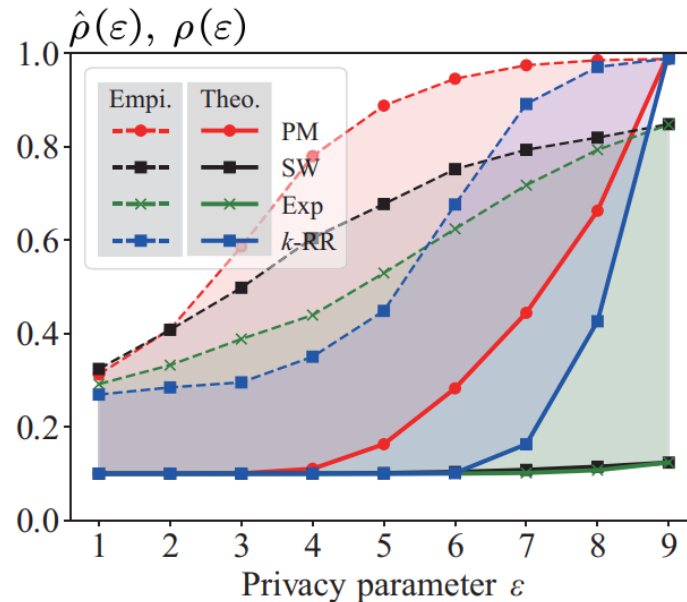
(a) RF: Average-case utility.



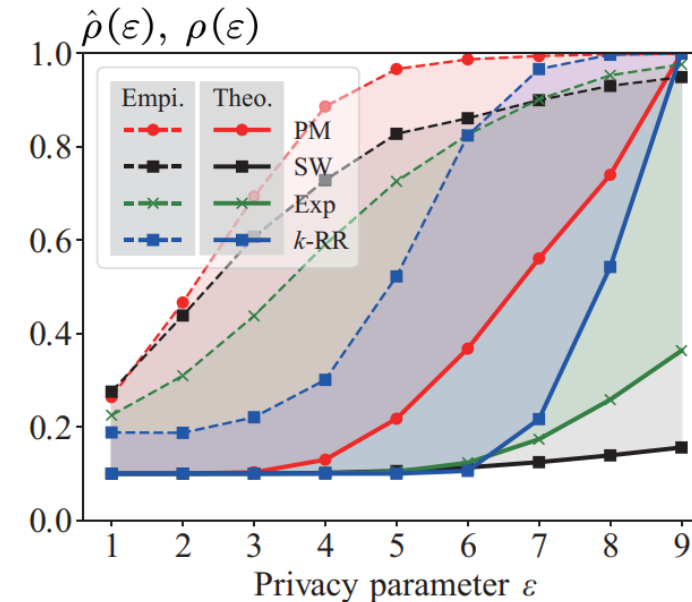
(b) RF: Worst-case utility.

Case Studies – Neural Networks (49-dim)

- Conservative analytical utility for high-dim classifiers
 - but high analytical utility $\rightarrow$ high empirical utility



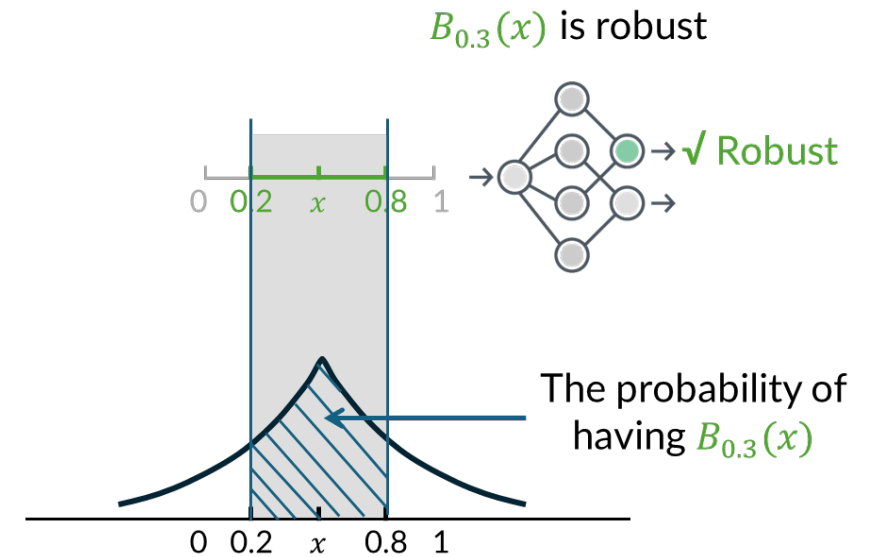
1st image



2nd image

Summary & Takeaway

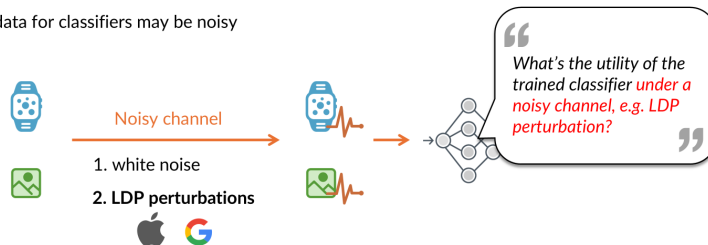
- **RQ: Classifier utility under LDP-perturbed inputs**
- **This paper:**
 - **connects LDP with robustness**
 - **provides analytical classifier utility**
 - the analytical utility is accurate for low-dim classifiers
 - the analytical utility is also useful for high-dim classifiers



Quantifying Classifier Utility under LDP

Classifier Utility under Noisy Inputs

- Input data for classifiers may be noisy



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For a better privacy-utility balance

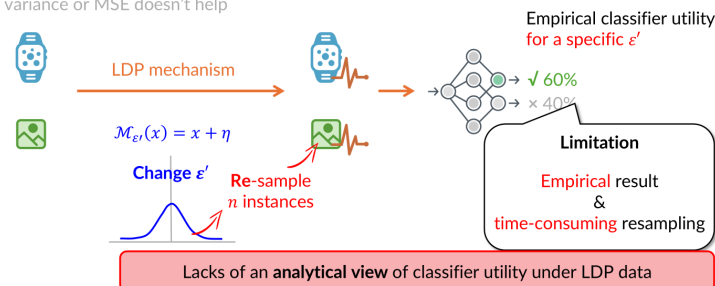
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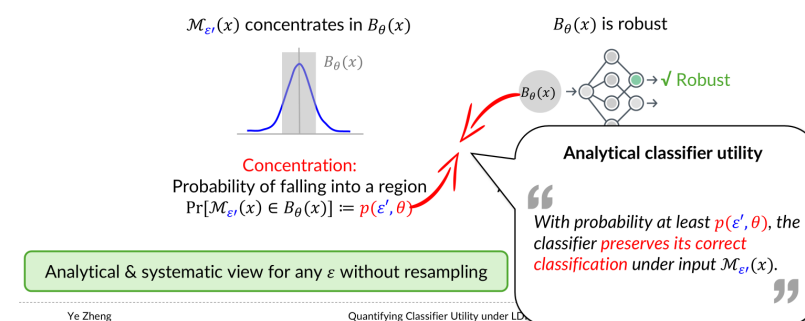
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Empirical Utility → Analytical Utility

- Analytical approach: connecting LDP with robustness

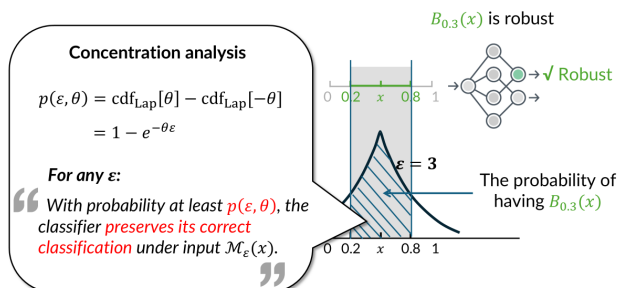


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Quantifying Classifier Utility under LDP

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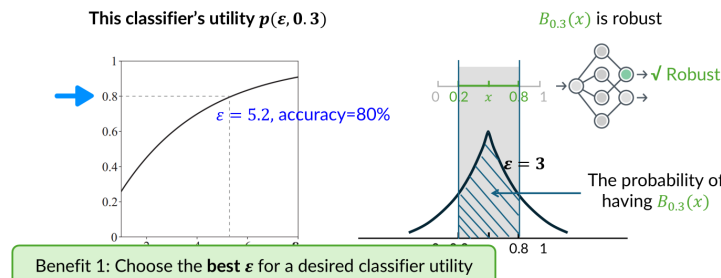
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Quantifying Classifier Utility under LDP

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Fixed Classifier (θ)

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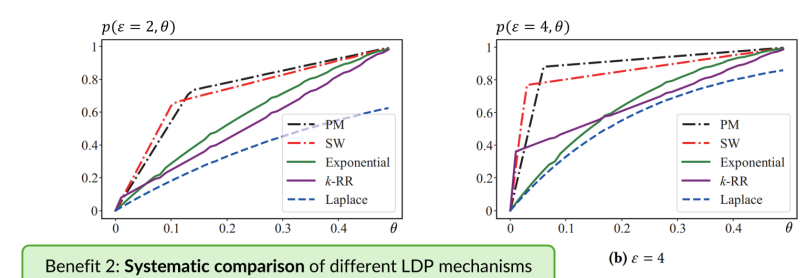
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Quantifying Classifier Utility under LDP

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Quantifying Classifier Utility under LDP

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Thank you!

